

Multi-class, unsupervised detection and classification of biological and anthropogenic sounds in coral reefs

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Abstract

Analyzing the complex and diverse soundscapes of ecosystems like coral reefs remains a challenge for understanding environmental dynamics and processes. While machine learning techniques can significantly improve detection and classification capabilities, applications of traditional supervised learning to underwater acoustics are limited by the size and class-coverage of labeled datasets. Unsupervised machine learning offers the potential to detect and classify sounds without the guidance of human labels, including signals that were unknown to the human operator. However, the majority of previously developed unsupervised approaches characterize reef soundscapes from correlative metrics without identifying specific sounds, and the few that detect individual signals have been trained on limited data (<10 days), which constrains the potential to generalize across datasets and geographical localities. Here, a convolutional autoencoder was built and trained on year-long acoustic datasets from four Hawaiian coral reefs, and latent embeddings were clustered using Gaussian mixture modeling. A total of 29 classes were automatically generated, and a manual review of samples in each class determined that nine of the classes corresponded to distinct biological and anthropogenic sounds. The classes were identified to be two call types from the damselfish, *Dascyllus*

albisella, parrotfish feeding sounds, holocentrid calls, an unidentified fish sound, three humpback whale song units, and ship noise. The classifier was found to be robust against an independently-collected dataset with *D. albisella* calls labeled (AUC=0.9). Diel, lunar, and seasonal trends were observed for all nine classes, including previously-unidentified responses of the holocentrid and unknown fish groups to lunar illumination. This work demonstrates the capability of unsupervised algorithms to cluster acoustic signals into identifiable biological and anthropogenic categories in order to examine and characterize ecological trends.

Author's Summary

Detection and classification of biological sounds is essential for monitoring and restoring coral reefs, yet most artificial intelligence (AI) methods need large, hand-labeled training datasets, which are difficult to obtain in acoustically complex reef environments. Here we present an unsupervised AI algorithm that automatically detects and clusters sounds from acoustic recordings with no manual annotation. We applied the algorithm to four concurrent, year-long recordings from Hawaiian coral reefs, and it automatically grouped detections into nine distinct signal types: two call types from Domino damselfish, feeding sounds from parrotfish, calls from holocentrids (squirrelfishes and soldierfishes), an unidentified fish sound, three separate humpback whale song segments, and ship noise. When tested on an independent dataset with labeled Domino damselfish calls, the model correctly identified more than 82% of damselfish calls while producing false positives on fewer than 15% of non-damselfish sounds. All nine classes exhibited patterns linked to sunlight or the seasons, and we discovered previously-unreported responses of the holocentrid and unknown fish groups to moonrise and lunar phase. This work demonstrates that unsupervised AI can reliably detect and cluster sound types and uncover ecological trends without the guidance of a human operator.

I. INTRODUCTION

Coral reefs are acoustically dynamic environments, with hundreds of diverse biological sounds often contained within a single minute of acoustic data [e.g., 1, 2, 3] alongside physical and anthropogenic noise [4, 5]. These soundscapes serve as important indicators of reef health, with acoustic diversity and activity shown to increase recruitment of pelagic fish, coral, and crustacean larva [2, 6-11], and to directly correlate with more resilient reefs [2, 12, 13]. However, to date, reef ecosystem function as inferred through soundscape metrics has largely been correlative, based on unidentified sound classes [12, 14, 15], band-limited sound levels [5, 16-19] or acoustic indices [2, 20-24], all of which lack taxonomic identification. By parsing multiple individual biological sounds and identifying them to a more precise taxonomic level, coral reef soundscapes can be interpreted to provide improved ecological context and actionable metrics for reef monitoring, conservation, and restoration. This is critical because natural resource management is primarily focused on individual species rather than sound production [3].

The application of machine learning detectors and classifiers allows for the identification of individual signals that would be missed by traditional intensity-based metrics and acoustic indices. For automated detection of fish acoustic signals, supervised machine learning has been typically applied to large, labeled datasets focusing on a limited set of call types, including calls from groupers [25, 26], damselfishes [27], toadfishes [28,29], and croakers [29]. While effective for identifying spatial or temporal detection patterns, these techniques are typically not transferable to other environments/locations, other sensors, or other call-types. Other supervised detectors have been trained on fish calls more broadly, without differentiating call-type or species [14, 15].

In contrast to supervised learning, unsupervised techniques enable rapid and inexpensive signal labeling without manual annotation. Unsupervised learning reduces inter- and intra-observer variability in event classification by circumventing subjective human interpretation, which can be

particularly useful for acoustically diverse datasets where individual sounds are unknown or difficult to disentangle. Previous unsupervised algorithms have been trained to identify biological choruses in long-term spectral averages with minute-scale temporal resolution [30-34], or to characterize soundscape recordings without identifying individual signals [23, 24]. While a number of studies have used unsupervised clustering to detect and identify sounds associated with marine mammals [35-39], fewer studies have used unsupervised techniques to classify fish calls. In a study by Noble et al. [40], handpicked spectral and temporal features were extracted from 55,000 manually labeled fish calls in order to cluster them into 55 unidentified sub-groups. A study by Ozanich et al. [41] differentiated fish and whale vocalizations using two unsupervised approaches: 1) clustering of handpicked spectral and temporal features similar to Noble et al. [40], and 2) clustering of deep latent features generated by convolutional autoencoders. More than 4,000 labeled detections were automatically clustered, and the deep learning approach was found to yield significant improvements in classification accuracy. The algorithms developed by Noble et al. [40] and Ozanich et al. [41] were both applied to less than ten days of data, which constrained the ability to resolve long-term diel or seasonal ecological trends.

Here, an unsupervised deep clustering algorithm was trained on 7.7 million acoustic signals over four concurrent, year-long datasets from Hawaiian coral reefs. A simple manual review of clusters identified nine classes corresponding to distinct signals, including two call types from the damselfish, *Dascyllus albisella*, soniferous signatures from holocentrids (squirrelfishes and soldierfishes), an unidentified fish call, scraping from parrotfish grazing, three song elements from humpback whales, and noise from ship traffic. The classifier was verified against an independently collected, human-annotated dataset containing labeled *D. albisella* calls with synchronized video-audio array verification (AUC=0.9). Seasonal and diel characteristics of well-studied signals (such as humpback whale and

damselfish calls) are consistent with known behaviors, while the temporal patterns of less-studied signals reveal previously-unidentified ecological trends, including the response of the holocentrid group to lunar illumination. This work demonstrates the capability of unsupervised algorithms to automatically detect and cluster soundscape features, including signals and patterns that were not known to the human analyst.

II. METHODS

Long-term passive acoustic measurements were collected at four survey sites with sensor depths ranging from 20-30 m off the western coast of Hawai'i Island (Fig 1). Survey sites were selected based on prior local knowledge to capture a wide range of coastal reef settings. HTI-96-min hydrophones were deployed with Loggerhead LS1 (Survey Sites 1 and 4) or Loggerhead LS1X (Survey Sites 2 and 3) recording packages. Acoustic recorders were set to record at a sampling rate of 96 kHz, recording for 1 minute every 15 minutes for the LS1 recorders and 1 minute every 10 minutes for the LS1X recorders to accommodate the 1-year deployment duration. The sensitivity of the HTI 96-min hydrophones used here was -170 dB re V/ μ Pa, and the frequency range was 2 Hz to 30 kHz. A synchronized video-audio array was independently deployed at a fifth location ("Test Site", Fig. 1) in order to provide ground-truth verification of calls from *Dascyllus albisella* (Domino damselfish). This is a well-documented vocalizing species in Hawaiian coral reefs [42-45], but few of their sounds are publicly available for model training. This passive acoustic camera combines an Insta-360 X2 360° camera with and within a tetrahedral hydrophone array and is described further in Dantzker et. al. [3] The passive acoustic camera was deployed near nests of *Dascyllus albisella* on July 27, 2022 and allowed for the opportunity to match sounds from specific focal fish species and their associated behaviors with vocalizations.

The data analyzed from the four Survey Sites spans from May 1, 2020 at 00:00 local time to May 1, 2021 at 00:00 local time. Each one-minute audio sample was downsampled to a sampling

frequency of 1600 Hz and run through a 4th-order highpass butterworth filter with critical frequency 30 Hz to eliminate low-frequency seismic noise. Spectrograms were generated with 32-sample fast-Fourier transforms with an overlap of 28. Frequencies between 150 and 750 Hz were isolated in order to target a frequency band of interest where diverse biological sounds are present [46].

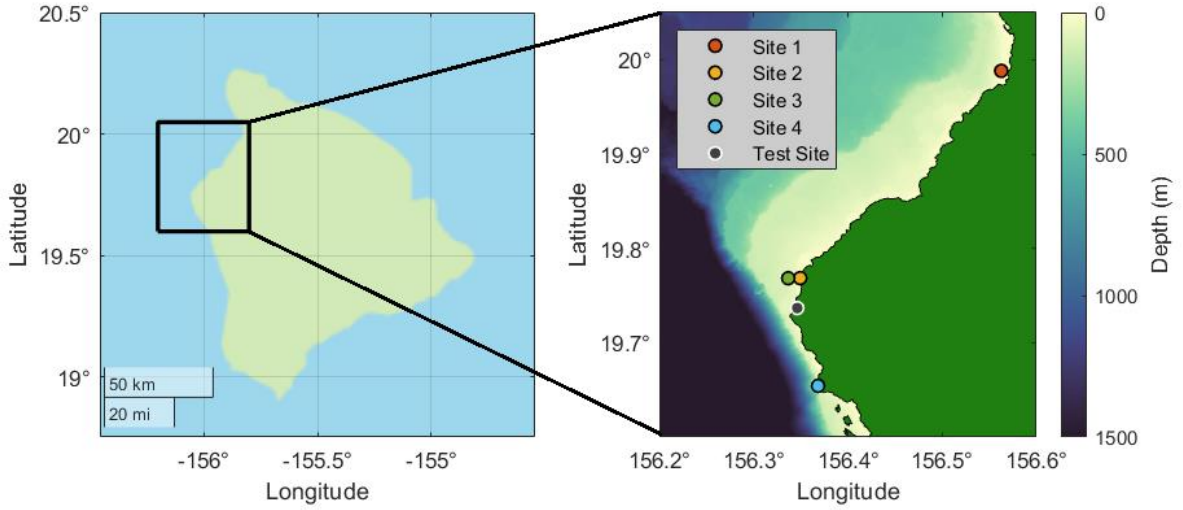


FIG. 1. Map of survey sites on the western coast of Hawai'i Island. Red, yellow, green, and blue dots correspond to long-term single-sensor survey sites, and the black dot corresponds to the test site with a synchronized video-audio array.

Detections were made using the Wang & Willett [47] power-law detector , where the power law statistic $PL(t)$ is defined as

$$PL(t) = \int_f \left(\frac{P_{xx}(t,f)}{B(t,f)} \right)^\nu df \quad (1)$$

where $P_{xx}(t,f)$ is the power spectral density in linear scale at time t and frequency f , $B(t,f)$ is the background noise level, and $\nu=1.7$ [47] . Here $B(t,f)$ is calculated for each frequency band as the 3-second median of $P_{xx}(t,f)$ centered at time t . The power law statistic was run through a time-domain Gaussian filter with standard deviation 10, and peaks were chosen at a threshold of 1

136 standard deviation above the one-minute mean. This low detection threshold was intentionally
 137 chosen so that the automated clustering algorithm will be trained to distinguish relevant signals from
 138 background or low signal to noise ratio (SNR) samples. For each detection, a spectrogram sample
 139 (PSD_{sample}) centered at the detection time was generated with size 12x144, corresponding to
 140 frequency range 150-750 Hz and duration 0.36 s. Spectrogram samples were standardized according
 141 to

$$142 \quad PSD' = \frac{PSD_{sample} - \mu_{sample}}{\sigma_{sample}} \quad (2)$$

143 where μ_{sample} and σ_{sample} are the mean and standard deviation of PSD_{sample} across both time
 144 and frequency bins. This standardization step ensures that signals with similar temporal/spectral
 145 characteristics but different intensities will be clustered similarly. Normalized samples are then
 146 obtained by clipping PSD' according to

$$147 \quad PSD_{norm} = \begin{cases} 0 & PSD' > 2 \\ PSD' - 1 & 1 < PSD' \leq 2 \\ 1 & PSD' \leq 1 \end{cases} \quad (3)$$

148 effectively setting the lower threshold at one standard deviation above the within-sample mean and
 149 the upper threshold at two standard deviations above the within-sample mean. Samples are clipped
 150 in this way in order to zero out background fluctuations and emphasize higher-SNR features. The
 151 detector was run on the full year of data in all four survey sites. In order to equalize detection rates
 152 between sites with different duty cycles, we discarded every third detection at Survey Sites 2 and 3,
 153 which had 1.5 times greater temporal coverage than Sites 1 and 4.

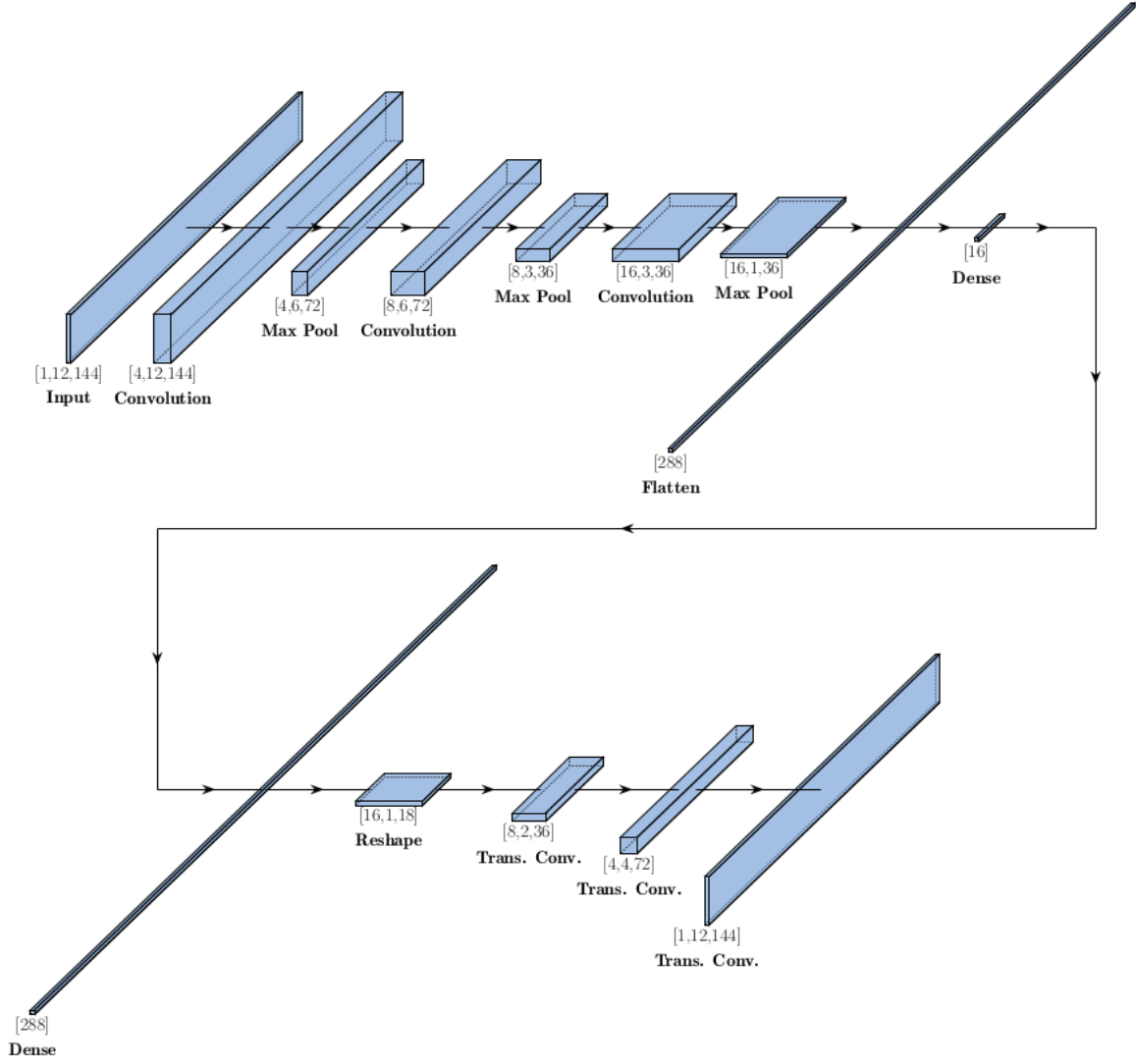


FIG. 2. Architecture of the convolutional autoencoder. The encoder compresses a 12x144 input into a 16 element latent embedding (top row). The decoder constructs a recreation of the input using only the latent embedding (bottom row).

TABLE I. Autoencoder architecture

Layer	Input Shape	Output Shape	Kernel Size	Stride	Padding	Filter	Activation
Convolution	[1,12,144]	[4,12,144]	[3,3]	[1,1]	[1,1]	4	ReLu

Max Pool	[4,12,144]	[4,6,72]	[2,2]	[2,2]	[0,0]	--	--
Convolution	[4,6,72]	[8,6,72]	[3,3]	[1,1]	[1,1]	8	ReLU
Max Pool	[8,6,72]	[8,3,36]	[2,2]	[2,2]	[0,0]	--	--
Convolution	[8,3,36]	[16,3,36]	[3,3]	[1,1]	[1,1]	16	ReLU
Max Pool	[16,3,36]	[16,1,18]	[2,2]	[2,2]	[0,0]	--	--
Reshape	[16,1,18]	[1,1,288]	--	--	--	--	--
Fully Connected	[1,1,288]	[1,1,16]	--	--	--	--	ReLU
Fully Connected	[1,1,16]	[1,1,288]	--	--	--	--	ReLU
Reshape	[1,1,288]	[16,1,18]	--	--	--	--	--
Trans. Conv.	[16,1,18]	[8,2,36]	[2,2]	[2,2]	[0,0]	16	ReLU
Trans. Conv.	[8,2,36]	[4,4,72]	[2,2]	[2,2]	[0,0]	8	ReLU
Trans. Conv.	[4,4,72]	[1,12,144]	[3,2]	[3,2]	[0,0]	4	ReLU

159

160 A total of 7,767,943 detections were made across the four survey sites (1,514,293 in Site 1,
161 1,933,200 in Site 2, 1,931,952 in Site 3, and 2,388,498 in Site 4). The detections were randomly split
162 into a training set (90%) and a held-out validation set (10%) and then fed into a convolutional
163 autoencoder which compresses inputs into a 16-dimensional latent space before reconstruction (Fig.
164 2, Table I). Training utilized the Adam optimizer with a learning rate of 0.0001 and a batch size of
165 64, minimizing the mean squared error (MSE) between input and output tensors. Training
166 concluded automatically when the epoch-to-epoch MSE reduction fell below 1×10^{-5} , requiring 35
167 epochs to reach convergence on the training set.

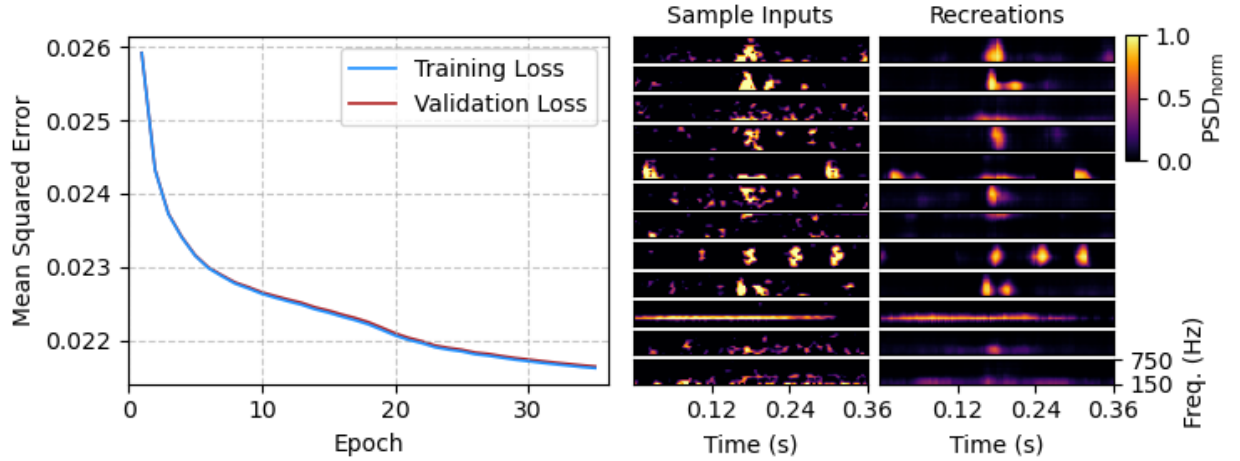


FIG. 2. Training/validation loss and sample recreation. Training and validation loss (left) is nearly identical after training ends, indicating that the autoencoder is not overfitting to the training set. Randomly-selected sample input spectrograms (middle) and reconstructions (right) show the autoencoder is retaining the basic spectral and temporal features in the latent embeddings.

After training, the encoder reprocessed the entirety of the dataset to extract latent feature representations. These representations were clustered using Gaussian mixture models (GMMs) with full covariance matrices, k-means++ initialization, and 10 random restarts, evaluating cluster counts from 10 to 50. For each candidate cluster count k , we identified “well-defined” clusters—those where $>10\%$ of assigned samples had likelihoods >0.99 —and computed the Akaike Information Criterion (AIC) exclusively using samples within these well-defined clusters. The optimal model ($k=29$) minimized this modified AIC. To interpret the acoustic content of these unsupervised groupings, we manually inspected 100 high-likelihood samples (likelihood > 0.99) per cluster by: 1) viewing their normalized spectrograms, and 2) listening to their raw audio. This process characterized the dominant acoustic signals captured by each cluster. When possible, clusters were associated to source-specific identified labels (e.g., species or vessel) based on sounds documented in the literature [e.g., 46, 48, 49] or focal observations confirmed with video [3].

III. RESULTS

Manual inspection of automatically generated clusters revealed that nine of the 29 clusters were associated with identifiable manmade and biological signals. These included two distinct damselfish sounds produced by *Dascyllus albisella* (“damselfish1,2”), scraping sounds from parrotfish feeding on coral (“parrotfish”), sounds from holocentrids (“holocentrid”), an unidentified fish sound (“unknown”), three distinct humpback whale song segments (“humpback1,2,3”), and ship noise (“ship”). Sounds from holocentrid fishes were inferred based on the similarity to sounds produced by *Myripristis berndti* and other members of the family [50-52], but we did not feel confident about assigning species-level labels to the holocentrid class. Example spectrograms are shown in Fig 3.

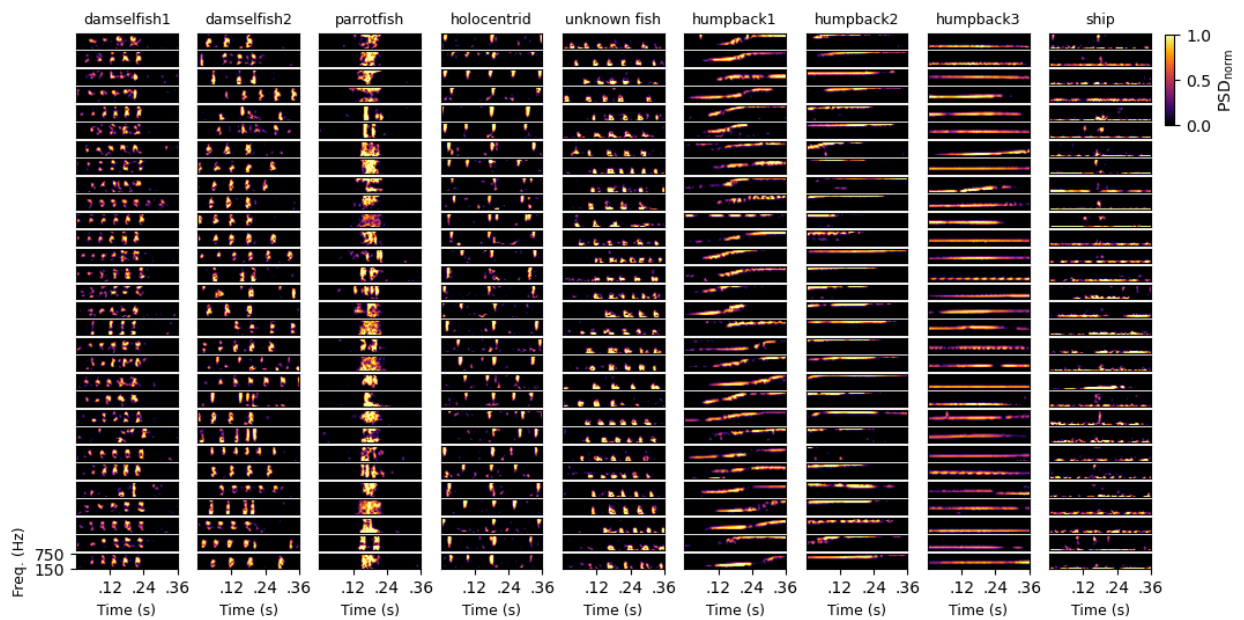


FIG. 3. Randomly selected spectrograms from “high-likelihood” ($L > 0.999$) samples for each of the 9 automatically-generated classes. Audio files containing sample sounds from each class are in S1 Audio.

While the duration and frequency range of spectrogram samples were specifically chosen to fit the characteristics of fish vocalizations (0.36 s duration, 150-750 Hz frequency range), the detector

also classified signals with durations and frequencies larger than the window ranges provided. Parrotfish scrapes have peak frequencies on the order of 2-4 kHz, but are broadband (200-8000 Hz) and commonly occupy into the lower frequencies studied here [53, 54]. Humpback song units typically have peak frequencies between 0-1.5 kHz, with song unit durations on the order of 1-3 s [e.g., 49, 55]. Tonals from passing ships can last minutes or hours, depending on the source level, speed and distance of the ship [e.g., 48, 56].

The distribution of samples with likelihoods greater than 0.99 are shown in Fig. 4. Significant variations in detection rates for each class are seen across the survey sites. Survey Site 1 saw a majority of damselfish2 call detections (51%) and a plurality of detections across all three humpback classes (40%, 39%, and 52%, respectively), as well as a strikingly low number of unknown fish detections (<0.5%). Site 2 saw a majority of unknown fish detections (67%) and a plurality of damselfish1 detections (46%). Site 3 saw a plurality of holocentrid (48%) and parrotfish (35%) detections. Site 4 was dominated by anthropogenic noise (47% of ship detections) with comparatively few biological detections (3.4% of humpback detections, 5.1% of holocentrid detections, and <15% for all other biological classes). The number of detections per class ranged from 20,000-50,000, with the exception of the damselfish1 class, where there were approximately 5,000.

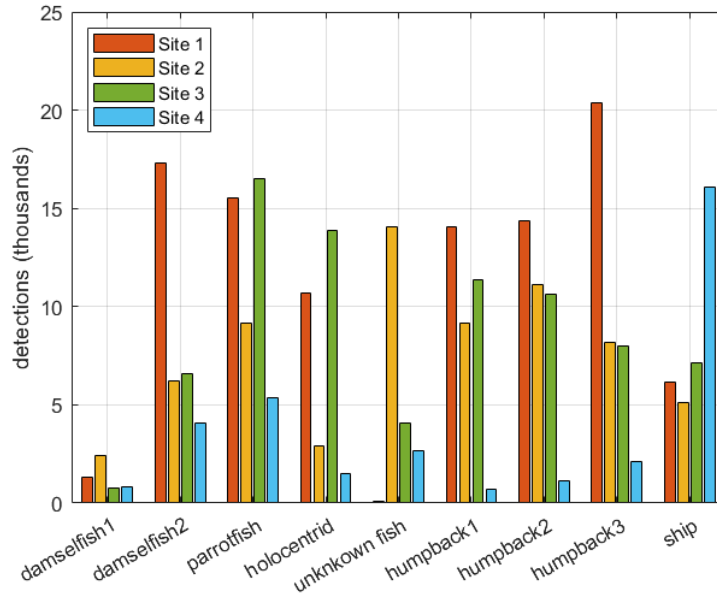


FIG. 4. Number of detections in each class ($L > 0.99$) across the 12 month recording period for the four survey sites.

Our classifier's damselfish1 and 2 classes were tested against the labeled dataset of verified *Dascyllus albisella* calls collected at the Test Site. The passive acoustic camera was deployed near a nest of *Dascyllus albisella* on July 27, 2022, and manual detections were made over a total of 57 minutes spanning 11:32-11:59 and 12:57-13:27 local time. 85 manual labels of the *Dascyllus* calls were verified by collocating beamformed acoustic detections with images of the individual from the 360° video imagery (Fig. 5a). All of these sounds were associated with *Dascyllus* courtship/territorial displays, and video samples for four of the calls can be found at <https://www.fisheycollaborative.org/fish-sounds/dascyllus-albisella>. The entire 57-minute dataset was run through the detection mechanism outlined in Section I, where 6,098 initial detections were made, 82 of which overlapped with the manual labels. The 6,098 detections were classified using the pre-trained encoder and GMM clustering algorithms, with no extra training on the labeled dataset. Detections were classified as damselfish when $L_1 + L_2 > T$ where L_1 and L_2 are the likelihood scores for the damselfish1 and

damselfish2 class assigned by the Gaussian mixture model, and T is an adjustable detection threshold. A ROC curve is generated by varying T from 0 to 1 (Fig. 5b). True positive rates of >82.5% are possible with false positive rates of <15%, and the area under the curve is 0.9.

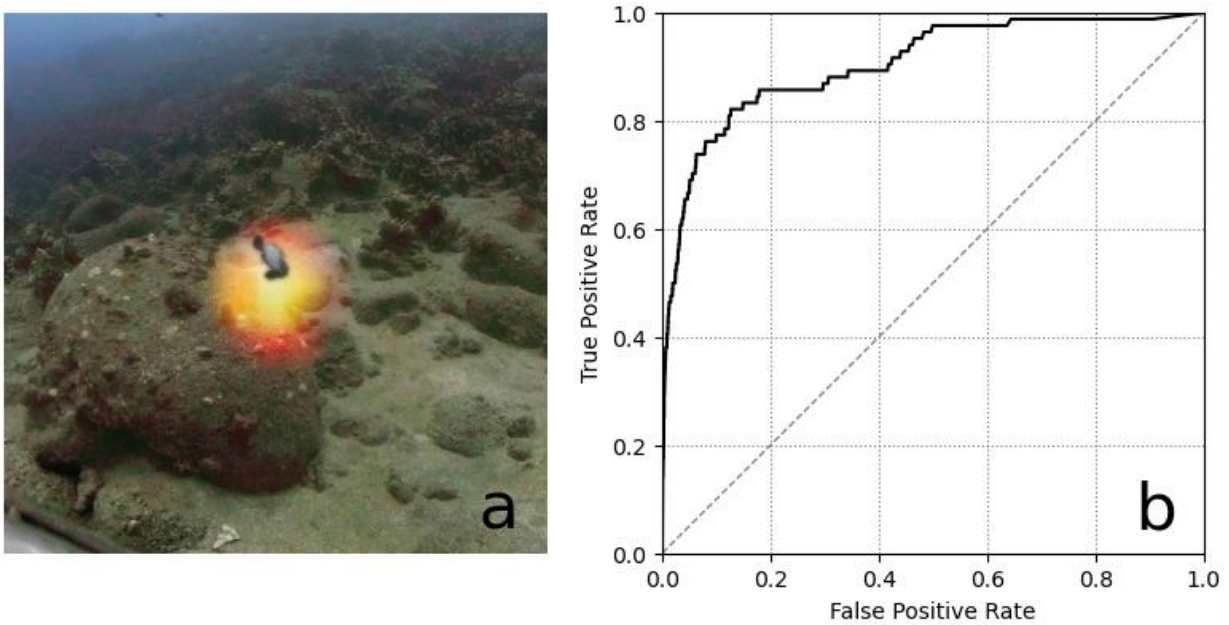


FIG. 5. Evaluating the unsupervised classifier with an independent, labeled dataset. The classifier's damselfish1 and 2 classes were tested against a 1-hour test dataset where a collocated acoustic array and 360° video system allowed for the visual identification of fish calls. A sample video frame is shown in (a), with an overlain visualization of the sound field energy distribution corresponding to the position of a *Dascyllus albisella*. A Receiver Operating Characteristic (ROC) curve for damselfish detections is shown in (b), with an area under the curve of 0.9. The visualization of the sound field energy is masked to make the individual visible in this illustration.

The diel and seasonal characteristics of the signals in each class are visualized in Fig. 6. For classes where diel and seasonal characteristics are known, the results of our classifier are consistent with previous observations. Humpback whale vocalizations occur almost exclusively between the

months of January and March, consistent with the known migration patterns of humpback whales to Hawaiian breeding grounds during the winter [57, 58]. Damselfish calls and parrotfish scrapes occur during the day, consistent with previous acoustic observations [27, 53]. Ship noise occurred at sporadic intervals mostly during daytime hours, which can be explained by the prevalence of recreational vessels associated with day trips rather than cargo vessels or cruise ships at these survey sites. Significant increases in “ship detections” occurred during the nights of February 5 and 15, and manual inspection of samples confirmed that increased seismic activity that bled into the >150 Hz frequency range were misclassified here as ship noise.

The results of the classifier also yielded insight into the temporal patterns of signals that have not been extensively studied. In particular, the holocentrid and unidentified calls display strong diel and lunar patterns. The unknown fish call occurs almost exclusively at night, during hours where the moon is not in the sky (diagonal cyan and yellow lines, Fig. 6). While changes in sound pressure level associated with moonrise have indicated a biological response to moonlight in this dataset [17], the automated detector and classifier presented here allowed us to isolate the specific signals responsible for the lunar chorus. The holocentrid call is found to be prominent in the 1-2 hours after sunset and before sunrise, with increases in activity during the new moon (white dots, Fig. 6).

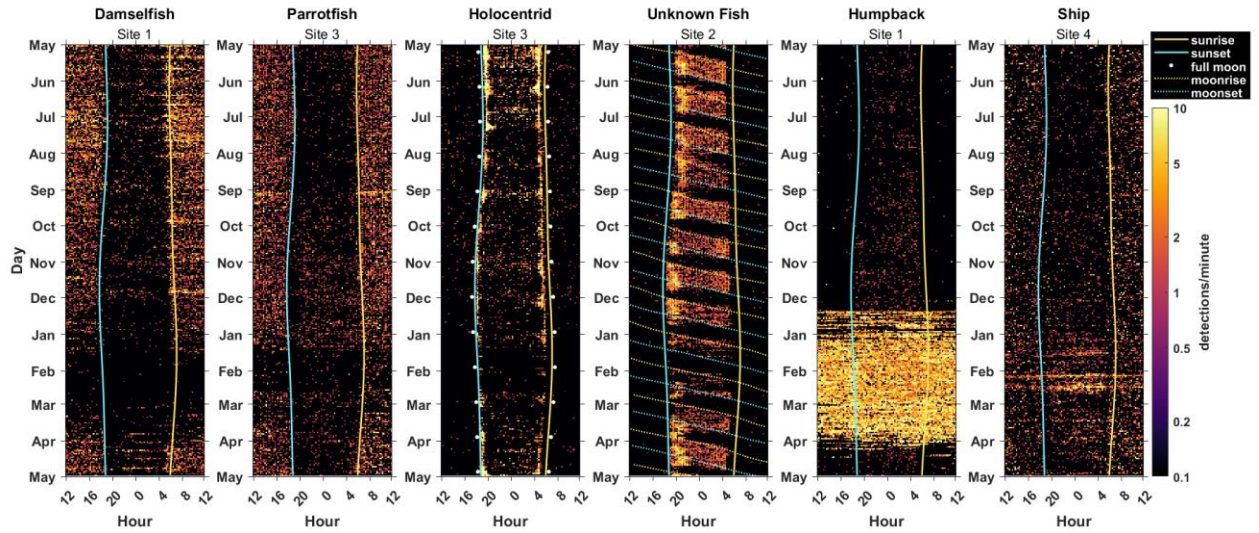


FIG. 6. Unique diel, lunar, or seasonal trends are observed for all classes. Detection rates are shown as a function of time of year (vertical axis) and time of day (horizontal axis) for classes at selected survey sites, where the two damselfish call types and the three humpback song units were merged into singular classes. Solid yellow and cyan lines respectively correspond to sunrise and sunset, dashed yellow and cyan lines respectively correspond to moonrise and moonset, and white dots correspond to full moon. Similar plots for all survey sites are shown in S1 Fig.

IV. DISCUSSION AND CONCLUSION

We demonstrated the ability to automatically detect and cluster signals that comprise underwater soundscapes with no manual labels. After training on four concurrent year-long datasets from Hawaiian coral reefs, nine clusters were automatically generated, including five distinct fish call classes, three humpback whale song units, and a ship noise class. The diel and seasonal patterns of damselfish, parrotfish, and humpback whale detections are consistent with previous literature, with damselfish and parrotfish active in the daytime [27, 53] and humpback songs detected in the winter months [57]. The holocentrid and unknown signals have not been identified previously, likely

because they occur during the nighttime when diver or camera surveys are not viable. The holocentrid and unknown signals exhibit distinct responses to sunrise, sunset, and lunar illumination, demonstrating the capability of unsupervised detection and classification algorithms to uncover new signals and trends in ambient soundscapes.

The classifier was found to be robust against a manually annotated *Dascyllus albisella* dataset (AUC=0.9), demonstrating the applicability of this model to other survey sites. While this model was specifically tailored to Hawaiian reefs, it would be straightforward to re-train the model on other acoustic datasets in order to extract commonly occurring signals in those environments. The duration and frequency range of training samples were chosen to reflect the general characteristics of fish vocalizations in reefs (0.36 seconds, 150-750 Hz), however the model was found to successfully characterize signals with longer durations (ship noise), larger frequency ranges (parrotfish scrapes), or both (humpback whale song units). The model architecture can therefore be applied with minimal prior knowledge of the signals that comprise the soundscape and can be used to potentially discover signals that were unknown to the human operator. If the temporal and spectral characteristics of a signal of interest are known, the duration and frequency range of input samples can be set accordingly. Classification accuracy can be further optimized using “human in the loop” approaches [1], where automatically generated clusters are manually refined or subdivided for supervised training. Further, while *D. albisella* sounds are well-characterized and readily identifiable [42, 43], the ability to extract identified sounds from a combined audio-video sensor array [3] highlights the opportunity to identify sound sources from the field, and then extract a large number of those species-labelled sounds for rapid retraining of machine learning models.

These results demonstrate the significant advantages offered by unsupervised machine learning for soundscape characterization and signal classification. Acoustic signals in a long-term dataset can be ingested, parsed, and automatically clustered without the burden and expense of manual labeling.

In contrast to traditional supervised learning approaches, the detection and clustering architecture introduced here can be easily applied to a variety of unlabeled acoustic datasets. These methods can be particularly useful for datasets where individual sounds are complex or difficult to disentangle, or where the specific signals that comprise the underwater soundscape have not yet been characterized. Such computational efforts will be essential for providing actionable, taxonomically-specific sound identification to enable passive acoustic monitoring for management and conservation of coral reef fishes and ecosystems at global scales [6, 59, 60].

V. SUPPORTING INFORMATION

S1 Audio. Audio samples for each of the nine automatically-generated classes. Audio files are produced for each class by collating ten randomly-selected one-second sound samples with a “high-likelihood” ($L > 0.999$) of belonging to that class.

S1 Fig. Diel and seasonal trends are for all classes and survey sites. Detection rates are shown as a function of time of year (vertical axis) and time of day (horizontal axis). Solid yellow and cyan lines respectively correspond to sunrise and sunset.

VI. ACKNOWLEDGMENTS

We would like to thank Justin Snow at Pacific Solutions, Gavin Key and Katie Key at Pacific Watersports, and David Mann at Loggerhead Instruments.

REFERENCES

1. Ferguson SR, Jensen FH, Hyer MD, Noble A, Apprill A, Mooney TA. Ground-truthing daily and lunar patterns of coral reef fish call rates on a US Virgin Island reef. *Aquat Biol.* 2022;31:77-87. <https://doi.org/10.3354/ab00755>
2. Kaplan MB, Mooney TA, Partan J, Solow AR. Coral reef species assemblages are associated with ambient soundscapes. *Mar Ecol-Prog Ser.* 2015;533:93-107. <https://doi.org/10.3354/meps11382>

- 328 3. Dantzker MS, Duggan MT, Berlik E, Delikaris-Manias S, Bountourakis V, Pulkki V, et al.
329 Deciphering complex coral reef soundscapes with spatial audio and 360° video. *Methods Ecol Evol*.
330 In press.
- 331 4. Ferrier-Pagès C, Leal MC, Calado R, Schmid DW, Bertucci F, Lecchini D, et al. Noise
332 pollution on coral reefs? — A yet underestimated threat to coral reef communities. *Mar Pollut Bull*.
333 2021;165:112129. <https://doi.org/10.1016/j.marpolbul.2021.112129>
- 334 5. Kaplan MB, Lammers MO, Zang E, Mooney TA. Acoustic and biological trends on coral
335 reefs off Maui, Hawaii. *Coral Reefs*. 2018;37(1):121-33. <https://doi.org/10.1007/s00338-017-1638-x>
- 336 6. Apprill A, Girdhar Y, Mooney TA, Hansel CM, Long MH, Liu YQ, et al. Toward a new era
337 of coral reef monitoring. *Environ Sci Technol*. 2023;57(13):5117-24.
338 <https://doi.org/10.1021/acs.est.2c05369>
- 339 7. Azofeifa-Solano JC, Parsons MJG, Brooker R, McCauley R, Pygas D, Feeney W, et al.
340 Soundscape analysis reveals fine ecological differences among coral reef habitats. *Ecol Indic*.
341 2025;171:11. <https://doi.org/10.1016/j.ecolind.2025.113120>
- 342 8. Gordon TAC, Radford AN, Davidson IK, Barnes K, McCloskey K, Nedelec SL, et al.
343 Acoustic enrichment can enhance fish community development on degraded coral reef habitat. *Nat*
344 *Commun*. 2019;10:7. <https://doi.org/10.1038/s41467-019-13186-2>
- 345 9. Lamont TAC, Williams B, Chapuis L, Prasetya ME, Seraphim MJ, Harding HR, et al. The
346 sound of recovery: Coral reef restoration success is detectable in the soundscape. *J Appl Ecol*.
347 2022;59(3):742-56. <https://doi.org/10.1111/1365-2664.14089>
- 348 10. Suca JJ, Lillis A, Jones IT, Kaplan MB, Solow AR, Earl AD, et al. Variable and spatially
349 explicit response of fish larvae to the playback of local, continuous reef soundscapes. *Mar Ecol-Prog*
350 *Ser*. 2020;653:131-51. <https://doi.org/10.3354/meps13480>
- 351 11. Tolimieri N, Jeffs A, Montgomery JC. Ambient sound as a cue for navigation by the pelagic
352 larvae of reef fishes. *Mar Ecol-Prog Ser*. 2000;207:219-24. <https://doi.org/10.3354/meps207219>
- 353 12. Jarriel SD, Formel N, Ferguson SR, Jensen FH, Apprill A, Mooney TA. Unidentified fish
354 sounds as indicators of coral reef health and comparison to other acoustic methods. *Front Remote*
355 *Sens*. 2024;5:1338586. <https://doi.org/10.3389/frsen.2024.1338586>
- 356 13. Freeman LA, Freeman SE. Rapidly obtained ecosystem indicators from coral reef
357 soundscapes. *Mar Ecol-Prog Ser*. 2016;561:69-82. <https://doi.org/10.3354/meps11938>
- 358 14. McCammon S, Formel N, Jarriel S, Mooney TA. Rapid detection of fish calls within diverse
359 coral reef soundscapes using a convolutional neural network. *J Acoust Soc Am*. 2025;157(3):1665-
360 83. <https://doi.org/10.1121/10.0035829>
- 361 15. Mouy X, Archer SK, Dosso S, Dudas S, English P, Foord C, et al. Automatic detection of
362 unidentified fish sounds: a comparison of traditional machine learning with deep learning. *Front*
363 *Remote Sens*. 2024;5:1439995. <https://doi.org/10.3389/frsen.2024.1439995>

- 364 16. Cato DH. Marine biological choruses observed in tropical waters near Australia. J Acoust
365 Soc Am. 1978;64(3):736-43. <https://doi.org/10.1121/1.382038>
- 366 17. Duane D, Freeman S, Freeman L. Moonlight-driven biological choruses in Hawaiian coral
367 reefs. PLoS One. 2024;19(3):e0299916. <https://doi.org/10.1371/journal.pone.0299916>
- 368 18. Nedelec SL, Simpson SD, Holderied M, Radford AN, Lecellier G, Radford C, et al.
369 Soundscapes and living communities in coral reefs: temporal and spatial variation. Mar Ecol-Prog
370 Ser. 2015;524:125-35. <https://doi.org/10.3354/meps11175>
- 371 19. Staatterman E, Rice AN, Mann DA, Paris CB. Soundscapes from a Tropical Eastern Pacific
372 reef and a Caribbean Sea reef. Coral Reefs. 2013;32(2):553-7. [https://doi.org/10.1007/s00338-012-](https://doi.org/10.1007/s00338-012-1007-8)
373 [1007-8](https://doi.org/10.1007/s00338-012-1007-8)
- 374 20. Bertucci F, Parmentier E, Lecellier G, Hawkins AD, Lecchini D. Acoustic indices provide
375 information on the status of coral reefs: an example from Moorea Island in the South Pacific. Sci
376 Rep. 2016;6:33326. <https://doi.org/10.1038/srep33326>
- 377 21. Dimoff SA, Halliday WD, Pine MK, Tietjen KL, Juanes F, Baum JK. The utility of different
378 acoustic indicators to describe biological sounds of a coral reef soundscape. Ecol Indic.
379 2021;124:107435. <https://doi.org/10.1016/j.ecolind.2021.107435>
- 380 22. Elise S, Urbina-Barreto I, Pinel R, Mahamadaly V, Bureau S, Penin L, et al. Assessing key
381 ecosystem functions through soundscapes: A new perspective from coral reefs. Ecol Indic.
382 2019;107:11. <https://doi.org/10.1016/j.ecolind.2019.105623>
- 383 23. Williams B, Balvanera SM, Sethi SS, Lamont TAC, Jompa J, Prasetya M, et al. Unlocking the
384 soundscape of coral reefs with artificial intelligence: pretrained networks and unsupervised learning
385 win out. PLoS Comput Biol. 2025;21(4):e1013029. <https://doi.org/10.1371/journal.pcbi.1013029>
- 386 24. Minier L, Rouch J, Sabbagh B, Bertucci F, Parmentier E, Lecchini D, et al. Visualization and
387 quantification of coral reef soundscapes using *CoralSoundExplorer* software. PLoS Comput Biol.
388 2025;21(4):36. <https://doi.org/10.1371/journal.pcbi.1012050>
- 389 25. Ibrahim AK, Chérubin LM, Zhuang HQ, Schärer Umpierre MT, Dalglish F, Erdol N, et al.
390 An approach for automatic classification of grouper vocalizations with passive acoustic monitoring.
391 J Acoust Soc Am. 2018;143(2):666-76. <https://doi.org/10.1121/1.5022281>
- 392 26. Ibrahim AK, Zhuang HQ, Schaerer-Umpierre M, Woodward C, Erdol N, Cherubin LM.
393 Fish Acoustic Detection Algorithm Research: a deep learning app for Caribbean grouper calls
394 detection and call types classification. Front Mar Sci. 2024;11:1378159
395 <https://doi.org/10.3389/fmars.2024.1378159>
- 396 27. Munger JE, Herrera DP, Haver SM, Waterhouse L, McKenna MF, Dziak RP, et al. Machine
397 learning analysis reveals relationship between pomacentrid calls and environmental cues. Mar Ecol-
398 Prog Ser. 2022;681:197-210. <https://doi.org/10.3354/meps13912>

- 399 28. Bohnenstiehl DR. Automated cataloging of oyster toadfish (*Opsanus tau*) boatwhistle calls
400 using template matching and machine learning. Ecol Inform. 2023;77:102268.
401 <https://doi.org/10.1016/j.ecoinf.2023.102268>
- 402 29. Huang Z, Ochs D, Amorim MCP, Fonseca PJ, Goel M, Nunes NJ, Vieira M, Lopes M.
403 Deep learning-based frameworks for the detection and classification of soniferous fish.. J Acoust
404 Soc Am. 2025;158(2):1060-1071. <https://doi.org/10.1121/10.0038800>
- 405 30. Kim EB, Frasier KE, McKenna MF, Kok ACM, Reeves LP, Oestreich WK, et al.
406 SoundScape learning: An automatic method for separating fish chorus in marine soundscapes. J
407 Acoust Soc Am. 2023;153(3):1710-22. <https://doi.org/10.1121/10.0017432>
- 408 31. Lin TH, Akamatsu T, Tsao Y. Sensing ecosystem dynamics via audio source separation: A
409 case study of marine soundscapes off northeastern Taiwan. PLoS Comput Biol. 2021;17(2):23.
410 <https://doi.org/10.1371/journal.pcbi.1008698>
- 411 32. Lin TH, Fang SH, Tsao Y. Improving biodiversity assessment via unsupervised separation of
412 biological sounds from long-duration recordings. Sci Rep. 2017;7:4547.
413 <https://doi.org/10.1038/s41598-017-04790-7>
- 414 33. Mahale VP, Chanda K, Chakraborty B, Salkar T, Sreekanth GB. Biodiversity assessment
415 using passive acoustic recordings from off-reef location-Unsupervised learning to classify fish
416 vocalization. J Acoust Soc Am. 2023;153(3):1534-53. <https://doi.org/10.1121/10.0017248>
- 417 34. Buchan SJ, Mahú R, Wuth J, Balcazar-Cabrera N, Gutierrez L, Neira S, et al. An
418 unsupervised Hidden Markov Model-based system for the detection and classification of blue whale
419 vocalizations off Chile. Bioacoustics. 2020;29(2):140-67.
420 <https://doi.org/10.1080/09524622.2018.1563758>
- 421 35. Frasier KE. A machine learning pipeline for classification of cetacean echolocation clicks in
422 large underwater acoustic datasets. PLoS Comput Biol. 2021;17(12):26.
423 <https://doi.org/10.1371/journal.pcbi.1009613>
- 424 36. Frasier KE, Roch MA, Soldevilla MS, Wiggins SM, Garrison LP, Hildebrand JA. Automated
425 classification of dolphin echolocation click types from the Gulf of Mexico. PLoS Comput Biol.
426 2017;13(12):23. <https://doi.org/10.1371/journal.pcbi.1005823>
- 427 37. Li K, Sidorovskaia NA, Tiemann CO. Model-based unsupervised clustering for
428 distinguishing Cuvier's and Gervais' beaked whales in acoustic data. Ecol Inform. 2020;58:101094.
429 <https://doi.org/10.1016/j.ecoinf.2020.101094>
- 430 38. Murray SO, Mercado E, Roitblat HL. The neural network classification of false killer whale
431 (*Pseudorca crassidens*) vocalizations. J Acoust Soc Am. 1998;104(6):3626-33.
432 <https://doi.org/10.1121/1.423945>
- 433 39. Cohen RE, Frasier KE, Baumann-Pickering S, Wiggins SM, Rafter MA, Baggett LM, et al.
434 Identification of Western North Atlantic odontocete echolocation click types using machine learning
435 and spatiotemporal correlates. PLoS One. 2022;17(3):e0264988.
436 <https://doi.org/10.1371/journal.pone.0264988>

40. Noble AE, Jensen FH, Jarriel SD, Aoki N, Ferguson SR, Hyer MD, et al. Unsupervised clustering reveals acoustic diversity and niche differentiation in pulsed calls from a coral reef ecosystem. *Front Remote Sens.* 2024;5:13. <https://doi.org/10.3389/frsen.2024.1429227>
41. Ozanich E, Thode A, Gerstoft P, Freeman LA, Freeman S. Deep embedded clustering of coral reef bioacoustics. *J Acoust Soc Am.* 2021;149(4):2587-601. <https://doi.org/10.1121/10.0004221>
42. Lobel PS, Mann DA. Spawning sounds of the damselfish, *Dascyllus albisella* (Pomacentridae), and relationship to male size. *Bioacoustics.* 1995;6:187-98.
43. Mann DA, Lobel PS. Passive acoustic detection of sounds produced by the damselfish, *Dascyllus albisella* (Pomacentridae). *Bioacoustics.* 1995;6:199-213. <https://doi.org/10.1080/09524622.1995.9753290>
44. Mann DA, Lobel PS. Acoustic behavior of the damselfish *Dascyllus albisella*: behavioral and geographic variation. *Environ Biol Fish.* 1998;51(4):421-8.
45. Laboury S, Parmentier E, Lobel PS. Are there individual acoustic signatures in the damselfish *Dascyllus albisella*? *J Acoust Soc Am.* 2025;157(1):48-56. <https://doi.org/10.1121/10.0034790>
46. Tricas TC, Boyle KS. Acoustic behaviors in Hawaiian coral reef fish communities. *Mar Ecol- Prog Ser.* 2014;511:1-16. <https://doi.org/10.3354/meps10930>
47. Wang Z, Willett PK. All-purpose and plug-in power-law detectors for transient signals. *IEEE Trans Signal Process.* 2001;49(11):2454-66. <https://doi.org/10.1109/78.960393>
48. McKenna MF, Rowell TJ, Margolina T, Baumann-Pickering S, Solsona-Berga A, Adams JD, et al. Understanding vessel noise across a network of marine protected areas. *Environ Monit Assess.* 2024;196(4):369. <https://doi.org/10.1007/s10661-024-12497-2>
49. Au WWL, Pack AA, Lammers MO, Herman LM, Deakos MH, Andrews K. Acoustic properties of humpback whale songs. *J Acoust Soc Am.* 2006;120(2):1103-10. <https://doi.org/10.1121/1.2211547>
50. Banse M, Bertimes E, Lecchini D, Donaldson TJ, Bertucci F, Parmentier E. Sounds as taxonomic indicators in Holocentrid fishes. *npj Biodiversity.* 2024;3(1):33. <https://doi.org/10.1038/s44185-024-00064-4>
51. Banse M, Hanssen N, Sabbe J, Lecchini D, Donaldson TJ, Iwankow G, et al. Same calls, different meanings: Acoustic communication of Holocentridae. *PLoS One.* 2024;19(11):e0312191. <https://doi.org/10.1371/journal.pone.0312191>
52. Salmon M. Acoustical behavior of the menpachi, *Myripristis berndti*, in Hawaii. *Pacific Science.* 1967;21:364-81.
53. Tricas TC, Boyle KS. Parrotfish soundscapes: implications for coral reef management. *Mar Ecol-Prog Ser.* 2021;666:149-69. <https://doi.org/10.3354/meps13679>

54. Sartori JD, Bright TJ. Hydrophonic study of the feeding activities of certain Bahamian parrot fishes, Family Scaridae. *Hydro-Lab J.* 1973;2(1):25-56.
55. Kügler A, Lammers MO, Zang EJ, Pack AA. Male humpback whale chorusing in Hawai'i and its relationship with whale abundance and density. *Front Mar Sci.* 2021;8:735664. <https://doi.org/10.3389/fmars.2021.735664>
56. Kline LR, DeAngelis AI, McBride C, Rodgers GG, Rowell TJ, Smith J, et al. Sleuthing with sound: Understanding vessel activity in marine protected areas using passive acoustic monitoring. *Mar Pol.* 2020;120:104138. <https://doi.org/10.1016/j.marpol.2020.104138>
57. Baker CS, Herman LM, Perry A, Lawton WS, Straley JM, Wolman AA, et al. Migratory movement and population structure of humpback whales (*Megaptera novaeangliae*) in the central and eastern North Pacific. *Mar Ecol-Prog Ser.* 1986;31(2):105-19. <https://doi.org/10.3354/meps031105>
58. Lammers MO, Goodwin B, Kügler A, Zang EJ, Harvey M, Margolina T, et al. The occurrence of humpback whales across the Hawaiian archipelago revealed by fixed and mobile acoustic monitoring. *Front Mar Sci.* 2023;10:15. <https://doi.org/10.3389/fmars.2023.1083583>
59. Mooney TA, Di Iorio L, Lammers M, Lin T-H, Nedelec SL, Parsons M, et al. Listening forward: Approaching marine biodiversity assessments using acoustic methods. *Royal Society Open Science.* 2020;7(8):201287.
60. Hodson EJ, Cox K, Juanes F, Looby A. Actively soniferous tropical reef fishes are diverse, vulnerable, and valuable. *Journal of Fish Biology.* 2025;106(4):990-5. <https://doi.org/10.1111/jfb.16030>